Audit Data Dictionary

analysis_confidence

Deterministic confidence label derived from analysis_reason_code. This is not the same as research_confidence. It reflects the pipeline's confidence in the current analysis classification.

Possible values, ordered by confidence:

- Blank = no analysis reason code is assigned.
- LOW = lower deterministic confidence, usually unresolved or less isolated diagnostics.
- MEDIUM = medium deterministic confidence, including Massive-focused diagnostics such as missing event adjustment, adjustment-factor continuity, partial/extra event capture, high-score anomaly, and methodology/source diagnostics such as event denominator or event source mismatch.
- HIGH = high deterministic confidence, such as close reversal, yFinance dividend/split factor mismatch, or yFinance event-date mismatch.

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analysis_reason_code

Deterministic diagnostic classification assigned by the audit pipeline. It explains the main reason the row needs review. When real-world research is available, the code may be adjusted so the diagnostic agrees with likely_correct_source.

Possible values, alphabetically:

- CLOSE_REVERSAL = Massive/yFinance return difference reverses on an adjacent trading day, suggesting a close-source or timing artifact rather than a corporate action.
  Example: On Tuesday, Massive return is -2.00% and yFinance return is -3.00%, so diff_return is +1.00%. On Wednesday, Massive return is +1.00% and yFinance return is +2.00%, so diff_return is -1.00%. The equal and opposite differences suggest a close timing issue, not a real event.
- EVENT_DENOMINATOR_MISMATCH = Massive and yFinance record the same dividend/split marker, but the event-return percentage differs because the same cash amount is divided by different prior-close values.
  Example: Massive reports cd:0.52, yFinance reports ca:0.52, and both sources show the same raw price return, but Massive's prior close is $28.00 while yFinance's prior close is $21.00. The same $0.52 cash dividend therefore produces different event-return percentages.
- EVENT_SOURCE_MISMATCH = Massive and yFinance report different dividend/split event markers for the date, especially when both sources have event markers but use different

event formatting, grouping, amounts, or source-event representation. When both sources have same-day event markers and the residual return difference is small, this is usually an event-representation issue; it can also be used as a broader fallback when event markers differ and pre-research evidence does not determine which source is economically correct.
Example: On the same ticker/date, Massive reports cd:0.25 while yFinance reports ca:0.50. With a $50.00 prior close, that is about +0.50% versus +1.00% expected dividend impact.

- HIGH_SCORE_ANOMALY = Massive return has a high heuristic anomaly score even when the Massive/yFinance return difference is not material.
  Example: Massive and yFinance both show +15.00% for the day, so diff_return is 0.00%. The ticker's recent daily moves are usually near 1.00%, so the +15.00% move still needs review.
- MS_ADJ_FACTOR_CONTINUITY = Rare fallback diagnostic for an unexplained Massive adjustment-chain discontinuity. This situation is expected to be uncommon and may not appear in ordinary audit runs; it exists as a guardrail for cases where Massive's cumulative adjustment-factor change does not align with the adjusted-return difference and the discrepancy is not better explained by a same-day event-source, event-denominator, missing-event, partial-event, or extra-event diagnosis.
  Example: Both sources show close rising from $100.00 to $101.00, a +1.00% price return. Both also show a $1.00 dividend, so the adjusted return should be about +2.00%. Massive's adjusted-return chain shows +1.20%, and no more specific event or denominator issue explains the difference.
- MS_EVENT_DATE_MISMATCH = Massive appears to have an event on the wrong trading date.
  Example: A $0.50 dividend has confirmed ex-date Tuesday. Massive shows cd:0.50 on Monday and no event on Tuesday, while yFinance shows ca:0.50 on Tuesday.
- MS_EXTRA_EVENT = Massive records an extra corporate-action event or extra event amount not supported by the comparison source and adjusted-return difference.
  Example: Research confirms one $0.65 dividend. Massive records cd:0.65 cd:0.65, yFinance records ca:0.65, and diff_return reconciles to Massive's extra $0.65 event-return impact.
- MS_MISSING_EVENT = Massive appears to be missing a corporate-action event or related adjustment needed to explain the return difference.
  Example: yFinance records sp:1.05 for a confirmed spin-off, implying about +5.00% adjusted-return impact. Massive has no event marker and shows a return about 5.00 percentage points lower.
- MS_PARTIAL_EVENT = Massive records a same-day corporate-action event, but the recorded event amount appears incomplete relative to the comparison source and the adjusted-return difference.
  Example: Research confirms a $0.15 dividend made up of a $0.075 base dividend plus a $0.075 variable dividend. Massive records cd:0.075, yFinance records ca:0.15, and

diff_return reconciles to the missing $0.075 event-return impact.

- MS_RETURN_METHOD_UNRESOLVED = Rare final fallback diagnostic for a Massive/yFinance return difference that remains unexplained after the specific event, denominator, date, close-reversal, adjustment-continuity, and dividend/split reconciliation checks have been applied. This situation is unexpected in ordinary audit runs; it exists as a guardrail for cases where the input fields do not isolate a single deterministic cause.

  Example: Massive return is +2.00% and yFinance return is +3.20%, so diff_return is -1.20%. There is no event marker mismatch, denominator mismatch, factor-continuity mismatch, dividend/split reconciliation break, or adjacent-day reversal to explain it.
- YF_DIV_SPLIT_RETURN_MISMATCH = yFinance implied dividend/split factor does not reconcile to yFinance explicit dividend/split factor.

  Example: yFinance records a $0.40 dividend and prior close is $40.00, so the explicit dividend impact is about +1.00%. yFinance's adjusted close implies a +2.50% dividend/split impact instead.
- YF_EVENT_DATE_MISMATCH = post-research override indicating yFinance appears to have the event on the wrong trading date.

  Example: Research confirms a $0.60 dividend belongs on Thursday. Massive shows cd:0.60 on Thursday, but yFinance shows ca:0.60 on Wednesday.
- YF_MISSING_EVENT = post-research override indicating yFinance appears to be missing a corporate-action event or related adjustment.

  Example: Research confirms a $0.75 dividend with prior close $75.00, so the expected impact is about +1.00%. Massive shows cd:0.75, but yFinance has no dividend marker or adjusted-return impact.

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date

Trading date being reconciled. This is the date of the Massive row and the date used to join yFinance prices, dividend/split event markers, return calculations, diagnostics, and optional real-world event research.

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diff_return

Material adjusted-return difference between Massive and yFinance, using the sign convention ms_return - yf_return. Positive means Massive's adjusted return is higher than yFinance's; negative means Massive's adjusted return is lower. Differences below the configured 1e-4 tolerance are set to null.

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event_bucket

External research classification for the identified real-world explanation.

Possible values, alphabetically:

- DISTRIBUTION = cash dividend, special dividend, return of capital, or similar cash/non-split distribution.
- MERGER = merger, acquisition, exchange offer, or transaction consideration affecting the price/return series.
- NEWS = material company, industry, macro, earnings, guidance, regulatory, litigation, or analyst/news event rather than a mechanical corporate action.
- PRICING_METHOD = a non-corporate-action difference caused by different vendor price, adjustment, or timing methods, after corporate actions and material news have been investigated.
- RIGHTS = rights offering, warrant distribution, subscription right, or similar shareholder right.
- SPINOFF = spin-off, split-off, or separation distribution.
- SPLIT = stock split, reverse split, or stock dividend treated as a split factor.
- UNKNOWN = no sufficiently supported real-world explanation was identified.

The analyst instructions require event precedence in this order: SPLIT, SPINOFF, DISTRIBUTION, MERGER, RIGHTS, NEWS, PRICING_METHOD, UNKNOWN.

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event_detected

External research field indicating whether a real-world event was identified for the ticker/date.

Possible values, alphabetically:

- Blank = no external real-world event research was joined.
- NO = research did not identify a relevant real-world event.
- UNCERTAIN = research did not support a firm yes/no conclusion.
- YES = research identified a relevant real-world event.

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evidence_summary

External research narrative summary. Per the analyst instructions, it should be row-specific, name Massive and yFinance explicitly, summarize sources reviewed, describe the real-world event or absence of one, explain economic plausibility, and tie the conclusion back to input row math such as ms_return, yf_return, diff_return, event markers, and expected return impact.

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expected_return_impact

External research estimate of the event's expected return impact using the audit's event-return convention. For splits, the instructions define this as split_factor - 1.0. For cash distributions, the approximate convention is cash_amount / prior_close. This value is used by real_world_events.apply_reason_overrides() to test whether the real-world event magnitude reconciles to signed diff_return within tolerance.

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has_div_split_mismatch

Boolean flag indicating whether Massive and yFinance report different dividend/split marker strings for the ticker/date. The comparison preserves source marker text in the output, but normalizes equivalent cash-action prefixes only while comparing: Massive cd and sc markers compare as generic ca cash-action markers against yFinance.

Possible values, alphabetically:

- false = Massive and yFinance dividend/split marker strings match after normalization.
- true = Massive and yFinance dividend/split marker strings differ after normalization.

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heuristic_anomaly_score

Deterministic pre-research anomaly score generated from Massive-side return behavior. The score is not a probability and does not identify the correct vendor by itself. It is a routing signal that helps identify rows where Massive's adjusted return looks unusual, even when Massive and yFinance returns agree.

Higher scores indicate that more independent anomaly signals were present.

Typical interpretation:

- 0 = no independent Massive-side anomaly signals were triggered. The row may still require review because of a vendor return difference, event-marker mismatch, or other reconciliation issue.
- 1-3 = low anomaly signal. One or more mild conditions may be present, but the row is usually reviewed only if another diagnostic also requires attention.
- 4-6 = moderate anomaly signal. The Massive return may be unusually large, inconsistent with nearby price behavior, or close to a corporate-action-style return pattern. These rows are worth review, especially when paired with event or adjustment differences.
- 7-10 = high anomaly signal. Multiple independent checks suggest the Massive return is unusual relative to price movement, adjustment mechanics, nearby returns, or corporate-action expectations.
- 11+ = very high anomaly signal. The row has several strong anomaly indicators and should be reviewed even if Massive and yFinance returns are identical, because both vendors may be reflecting a real market event or both may share a questionable treatment.

Important notes:

- The score is additive: multiple smaller signals can produce a high score.
- The score is pre-research only. Real-world research and input row math determine the final classification.
- A high score does not imply Massive is wrong. It means the row deserves investigation.
- A score of 0 does not imply the row is correct. It may still be flagged by return differences, event mismatches, adjustment-factor issues, or close-reversal diagnostics.

likely_correct_source

External research conclusion identifying which source appears economically correct after considering both real-world evidence and input row math. The analyst instructions require this to follow from ms_return, yf_return, diff_return, event markers, event-return fields, adjustment factors, and external evidence.

Possible values, alphabetically:
- Blank = no external real-world event research was joined.
- BOTH = Massive and yFinance are economically equivalent or both appear reasonable for the row.
- MASSIVE = Massive appears economically correct.
- NEITHER = neither Massive nor yFinance appears economically correct.
- UNCERTAIN = research cannot determine the likely correct source.
- YFINANCE = yFinance appears economically correct.

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massive_fix_action

Suggested Massive review or remediation action, generated from analysis_reason_code and optional real-world research. When research concludes likely_correct_source is MASSIVE or BOTH, this is cleared because no Massive remediation is recommended. Otherwise examples include adding or correcting missing corporate actions, reviewing adjustment-factor history or event denominators, rebuilding adjusted close and return chains, or manually reviewing close/adjusted-close/corporate-action inputs.

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massive_fix_priority

Priority for the suggested Massive remediation. This is research-aware: it is blank when real-world research concludes likely_correct_source is MASSIVE or BOTH.

Possible values, ordered by priority:
- Blank = no Massive remediation priority is assigned.
- LOW = low-priority Massive review, usually lower-score heuristic anomaly.
- MEDIUM = more investigative Massive review, such as an unresolved adjusted-return calculation issue or higher-score anomaly.
- HIGH = direct Massive fix candidate, such as missing event adjustment, event-date mismatch, adjustment-factor continuity, partial event capture, or extra event capture.

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massive_needs_fix

Boolean flag indicating whether Massive may require correction after deterministic diagnostics and optional real-world research. If research concludes likely_correct_source is MASSIVE or BOTH, this is false. If research concludes likely_correct_source is YFINANCE or NEITHER, this is true. Without research, it follows Massive-focused reason codes such as missing event adjustment, event-date mismatch, adjustment-factor continuity issue, partial event capture, extra event capture, unresolved Massive adjusted-return calculation issue, or high-score anomaly. Generic EVENT_SOURCE_MISMATCH rows require review but do not by themselves imply Massive needs a fix.

Possible values, alphabetically:
- false = Massive does not currently appear to need correction.
- true = Massive may require correction or manual remediation review.

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massive_problem_and_fix

Summary-report-only display column that vertically combines massive_problem_summary, massive_why_incorrect, and massive_fix_action with blank lines between the three sections.

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massive_problem_summary

Human-readable summary of the audit issue or suspected Massive-side problem, generated from analysis_reason_code and optional real-world research. Blank when the pipeline does not identify a review issue or when research concludes likely_correct_source is MASSIVE or BOTH.

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massive_why_incorrect

Human-readable explanation of why Massive may be incorrect or why the row needs Massive review/context, generated from analysis_reason_code and optional real-world research. Blank when the pipeline does not identify a review issue or when research concludes likely_correct_source is MASSIVE or BOTH.

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ms_adj_close

Massive adjusted close calculated by this audit pipeline as ms_close * ms_adj_factor. This is the Massive-side adjusted close used to calculate ms_return.

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ms_adj_factor

Massive cumulative backward-looking adjustment factor calculated by this audit pipeline from Massive split and dividend records. For splits, the factor uses split_from / split_to. For dividends, the factor uses (prior_close - cash_amount) / prior_close. The cumulative factor includes events after the row date so historical closes can be restated on a current adjusted basis.

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ms_close

Massive unadjusted close price for the ticker/date. This comes from Massive unadjusted OHLCV data and is used as the raw close input for Massive adjusted-close and raw price-return calculations.

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ms_div_split

Compact Massive corporate-action marker string for the ticker/date. Split events are represented as sp:<split_factor>, where split_factor is split_to / split_from. Regular cash dividends are represented as cd:<cash_amount>. Special cash dividends are represented as sc:<cash_amount>. Generic cash distributions with no Massive regular/special distinction are represented as ca:<cash_amount>. Multiple same-day events are space-separated.

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ms_return

Final Massive adjusted return for the ticker/date, calculated from the percentage change in the pipeline's Massive adjusted close series.

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ms_return_price

Massive raw price return from unadjusted closes, calculated as the percentage change from the prior Massive unadjusted close to the current Massive unadjusted close for the same ticker.

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needs_review

Boolean review flag. True when diff_return is non-null or when the Massive heuristic anomaly score is greater than or equal to MIN_SCORE_TO_REVIEW, which is currently 5.

Possible values, alphabetically:
- false = row does not meet review criteria.
- true = row meets review criteria.

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primary_source_url

Primary URL supporting the real-world event research conclusion. The instructions prefer authoritative sources such as SEC filings, investor relations releases, exchange notices, or official corporate-action announcements where available.

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real_world_event

External research description of the real-world event or explanation associated with the ticker/date. This may describe a split, distribution, merger, rights offering, news event, pricing-method explanation, or the absence/uncertainty of a real event depending on the analyst output.

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real_world_evidence

Summary-report-only display column that vertically combines real_world_event and evidence_summary with a blank line between the two sections. It gives the event headline followed by the supporting research and input row reconciliation.

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research_confidence

External research confidence label for the likely_correct_source and event conclusion. This is separate from analysis_confidence: research_confidence comes from the analyst/research evidence scoring rules, while analysis_confidence comes from the deterministic audit classification.

Possible values, ordered by confidence:
- Blank = no external real-world event research was joined.
- LOW = weak, ambiguous, or absence-of-event support.
- MEDIUM = reasonable but incomplete or partly inferential support.
- HIGH = strong source support and clear return/event reconciliation.

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review_priority

Review priority bucket. This is not the same as massive_fix_priority: review_priority records why the row merited review, while massive_fix_priority records whether Massive has a recommended fix after research.

Possible values, ordered by priority:
- Blank = no review is required.
- LOW = heuristic_anomaly_score is at least MIN_SCORE_TO_REVIEW but higher-priority conditions do not apply.
- MEDIUM = heuristic_anomaly_score is at least 8 and the row may be actionable.
- HIGH = diff_return is non-null.

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secondary_source_url

Secondary or confirming URL supporting the real-world event research conclusion. The instructions allow credible independent sources such as Reuters, Bloomberg, Yahoo Finance, Nasdaq, MarketWatch, Macrotrends, or company press releases.

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ticker

Normalized ticker symbol being audited. The pipeline strips surrounding whitespace and uppercases ticker symbols before joining Massive and yFinance data.

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yf_adj_close

yFinance adjusted close for the ticker/date, taken from yFinance OHLCV data. This is the adjusted close used to calculate yf_return.

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yf_adj_factor

Implied yFinance adjustment factor for the ticker/date, calculated as yf_adj_close / yf_close when yf_close is present and nonzero. Unlike ms_adj_factor, this is inferred from yFinance's supplied adjusted close rather than rebuilt directly from yFinance event records.

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yf_close

yFinance unadjusted close price for the ticker/date. This comes from yFinance OHLCV data and is used for yFinance raw price-return calculations and to infer the yFinance adjustment factor.

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yf_div_split

Compact yFinance corporate-action marker string for the ticker/date. Split events are represented as sp:<split_factor>, where split_factor is yFinance's split_ratio. Cash dividend/distribution events are represented as ca:<cash_amount> because yFinance does not expose the same Massive regular cash dividend cd versus special cash dividend sc distinction. Multiple same-day events are space-separated.

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yf_return

Final yFinance adjusted return for the ticker/date, calculated from the percentage change in yFinance adjusted close.

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yf_return_price

yFinance raw price return from unadjusted closes, calculated as the percentage change from the prior yFinance unadjusted close to the current yFinance unadjusted close for the same ticker.
